

Shubham Choudhury

PhD Candidate in Computational Biology | Postdoctoral Applicant

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RESEARCH INTERESTS

My research sits at the intersection of computational biology, RNA biology, and machine learning. I develop predictive tools and resources for the functional annotation of long non-coding RNAs (lncRNAs) and other RNA classes, with particular emphasis on subcellular localization, RNA-protein interactions, and their dysregulation in disease. I am keen to extend this work toward therapeutic applications — specifically RNA-based drug targets and biomarker discovery — by leveraging large language models trained on genomic sequences and integrating multi-omics data.

EDUCATION

Indraprastha Institute of Information Technology, Delhi (IIIT-D)

Jan 2021 – Present (Thesis submitted)

Doctor of Philosophy in Computational Biology | CGPA: 8.29 / 10.00

New Delhi, India

- **Thesis:** Development of AI-Based Tools for Functional Annotation of lncRNA and Its Applications
- **Supervisor:** Prof. Gajendra P.S. Raghava (Head, Computational Biology)
- Awarded the prestigious **DST-INSPIRE Fellowship** for doctoral research.

Indian Institute of Science Education and Research, Kolkata (IISER)

July 2014 – July 2019

BS-MS Dual Degree in Biological Sciences | CGPA: 7.11 / 10.00

Kolkata, India

- **Master's Thesis:** Transcriptomic Meta-Analysis of Human Sepsis and Related Illnesses
- Qualified **DBT-JRF (Category I)**, awarded by the Dept. of Biotechnology, Govt. of India.

RESEARCH EXPERIENCE

CSIR – Institute of Microbial Technology, Chandigarh (IMTECH)

July 2019 – July 2020

Project Assistant | Supervisor: Dr. Manoj Kumar

Chandigarh, India

- Built **CoronaVR**, a web resource consolidating epitope and therapeutic data for SARS-CoV-2 (*Frontiers in Microbiology*, 2020).
- Conducted metagenomic sequencing and virome analysis of sewage samples across Chandigarh to track environmental spread of viral pathogens.
- Performed computational drug repurposing for virus-associated diseases via drug-target network analysis, contributing to a publication in *Computers in Biology and Medicine* (2021).

KEY RESEARCH PROJECTS

lncrnaPI — LLM-Based Prediction of lncRNA-Protein Interacting Pairs 🌐

Nov 2025

Python · DNABERT-2 · ESM-2 · CatBoost · CNN · BLAST · Nfeature · Pfeature · POSSUM · Web Server Development 🔄

- Curated the largest non-redundant LPI dataset to date (524,488 pairs from NPInter v5.0); generated features using DNABERT-2, ESM-2-t30, PSSM profiles, and compositional descriptors.
- Best model (CatBoost + LLM embeddings) achieved **AUC = 0.989**, **MCC = 0.915**, substantially outperforming the only existing functional tool (LPI-BLS, AUC = 0.512).
- Deployed a web server and released lncrnaPI as a Python package on PyPI for genome-scale lncRNA-protein interaction scanning.

LLM-Based Prediction of Pancreatic Cancer from Blood Exosomal Transcriptomics

June 2024

Python · PeptideBERT · ProtBERT · ESM2 · BLAST · MERCI · Scikit-learn · PyTorch 🔄

- Mapped exosomal gene expression profiles from 501 PDAC/non-PDAC patients into protein sequence representations.
- Fine-tuned three protein language models (PeptideBERT, ProtBERT, ESM2) for binary cancer classification.
- Best model (ESM2-t6_8M_UR50D) achieved **AUROC = 0.95** on an independent held-out dataset.
- Developed an ensemble combining LLM-based predictions with alignment-based approaches (BLAST, MERCI).

CytoLNCpred — Subcellular Localization of lncRNA Across 15 Cell Lines 🌐

Jan 2023

Python · Machine Learning · DNABERT-2 · Web Server Development 🔄

- Curated lncRNA localization data from lncAtlas; extracted *k*-mer composition, correlation, and DNABERT-2 embedding features.
- Trained and benchmarked classical ML models and a transformer-based LLM; achieved **mean AUROC = 0.709** across 15 cell lines.
- Deployed a publicly accessible prediction web server; published in *Frontiers in Bioinformatics* (2025).

MRSLPred — Multi-Label Subcellular Localization of Human mRNA 🌐

Jan 2022

Python · Machine Learning · NFeature · Web Server Development 🔄

- Built a genome-scale predictor for six mRNA localization compartments using features from the RNALocate database.
- Boosted accuracy via MERCI motif features; achieved **mean AUROC = 0.742**. Published in *Frontiers in Bioinformatics* (2024).

Transcriptomic Meta-Analysis of Human Sepsis (Master's Thesis)

May 2018 – June 2019

R · RStudio · Bash · HPC Computing

- Analyzed public sepsis microarray datasets from GEO to identify differentially expressed genes and pathways.
- Assessed whether upregulation of the Osteoclast Differentiation Pathway is sepsis-specific or a generic inflammatory response.

TEACHING EXPERIENCE

Indraprastha Institute of Information Technology, Delhi (IIIT-D)

Winter 2022 – Winter 2024

Teaching Assistant, Department of Computational Biology

New Delhi, India

- BIO543: Big Data Mining in Healthcare (Winter 2022, Winter 2024)
- BIO542: Machine Learning for Biomedical Applications (Monsoon 2022)
- Assisted with tutorial sessions, assignment design, grading, and student mentoring for graduate-level courses in machine learning and biomedical data science.

TECHNICAL SKILLS

Programming & Scripting: Python, R, Bash, PHP

Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, Keras; Transformers/LLMs (DNABERT-2, ESM2, ProtBERT)

Bioinformatics Libraries: Pandas, NumPy, Bioconductor, Biopython

Databases & Web: MySQL, REST APIs, Apache, HTML5; experience deploying bioinformatics web servers

DevOps & HPC: Git, GitHub, Docker, HPC/SLURM cluster computing

Specialised Domains: lncRNA biology, RNA subcellular localization, RNA-protein interaction, environmental viromics, LLMs for genomics

Wet-Lab Techniques: RNA/DNA extraction, PCR, gel electrophoresis, viral enrichment, RNA-seq library preparation (Nanopore)

PUBLICATIONS

First-Author Publications

- [1] **S. Choudhury**, N. Bajiya, G.P.S. Raghava. "Prediction of lncRNA-protein interacting pairs using LLM embeddings based on evolutionary information." *bioRxiv* (2025). [Preprint; Under Review] [DOI]
- [2] **S. Choudhury**, N.K. Mehta, G.P.S. Raghava. "CytoLNCpred: A computational method for predicting cytoplasm-associated long non-coding RNAs in 15 cell lines." *Frontiers in Bioinformatics* 5 (2025): 1585794. [DOI]
- [3] **S. Choudhury**, A.S. Rathore, G.P.S. Raghava. "Compilation of resources on subcellular localization of lncRNA." *Frontiers in RNA Research* 2 (2024): 1419979. [DOI]
- [4] **S. Choudhury**, N. Bajiya, S. Patiyal, G.P.S. Raghava. "MRSLpred — a hybrid approach for predicting multi-label subcellular localization of mRNA at the genome scale." *Frontiers in Bioinformatics* 4 (2024): 1341479. [DOI]

Co-Authored Publications

- [5] S. Aggarwal, A. Dhall, S. Patiyal, **S. Choudhury**, A. Arora, G.P.S. Raghava. "An ensemble method for prediction of phage-based therapy against bacterial infections." *Frontiers in Microbiology* 14 (2023): 1148579. [DOI]
- [6] N. Bajiya, **S. Choudhury**, A. Dhall, G.P.S. Raghava. "AntiBP3: A method for predicting antibacterial peptides against Gram-positive/negative/variable bacteria." *Antibiotics* 13(2) (2024): 168. [DOI]
- [7] A. Dhall, S. Patiyal, **S. Choudhury**, S. Jain, K. Narang, G.P.S. Raghava. "TNFepitope: A webserver for the prediction of TNF- α inducing epitopes." *Computers in Biology and Medicine* 160 (2023): 106929. [DOI]
- [8] P.S. Gahlot, **S. Choudhury**, N. Bajiya, N. Kumar, G.P.S. Raghava. "Prediction of plant resistance proteins using alignment-based and alignment-free approaches." *Proteomics* 25(5–6) (2025): e202400261. [DOI]
- [9] A.K. Gupta, M.S. Khan, **S. Choudhury**, et al. "CoronaVR: A computational resource and analysis of epitopes and therapeutics for SARS-CoV-2." *Frontiers in Microbiology* 11 (2020): 1858. [DOI]
- [10] N. Kumar, **S. Choudhury**, N. Bajiya, S. Patiyal, G.P.S. Raghava. "Prediction of anti-freezing proteins from their evolutionary profile." *Proteomics* 25(3) (2025): e202400157. [DOI]
- [11] N. Kumar, S. Patiyal, **S. Choudhury**, R. Tomer, A. Dhall, G.P.S. Raghava. "DMPPred: A tool for identification of antigenic regions responsible for inducing Type 1 Diabetes Mellitus." *Briefings in Bioinformatics* 24(1) (2023): bbac525. [DOI]
- [12] A. Mukhopadhyay, **S. Choudhury**, M. Kumar. "Metaviromic analyses of DNA virus community from sediments of the N-Choe Stream, North India." *Virus Research* 330 (2023): 199110. [DOI]
- [13] A. Rajput, A. Thakur, A. Rastogi, **S. Choudhury**, M. Kumar. "Computational identification of repurposed drugs against viruses causing epidemics and pandemics via drug-target network analysis." *Computers in Biology and Medicine* 136 (2021): 104677. [DOI]

- [14] A.S. Rathore, **S. Choudhury**, A. Arora, P. Tijare, G.P.S. Raghava. "ToxinPred 3.0: An improved method for predicting the toxicity of peptides." *Computers in Biology and Medicine* **179** (2024): 108926. [DOI]
- [15] A.S. Rathore, N. Kumar, **S. Choudhury**, N.K. Mehta, G.P.S. Raghava. "Prediction of hemolytic peptides and their hemolytic concentration." *Communications Biology* **8**(1) (2025): 176. [DOI]

COMPETITIONS AND CHALLENGES

CODA TB DREAM Challenge — 3rd Place (International Team Competition)

March 2024

Machine Learning · Audio Signal Processing · Tuberculosis Screening · Digital Health

- Collaborated with lab members in the international **CODA TB DREAM Challenge**, focused on AI-based tuberculosis screening using cough audio recordings and clinical metadata.
- Developed predictive machine learning models for identifying TB-positive individuals from elicited cough sounds collected across 7 countries.
- Secured **3rd position globally** among participating teams in the challenge organized under the DREAM Challenges framework.
- Challenge findings and benchmark results were published in:
"CODA TB DREAM Challenge: Predicting Tuberculosis from Cough Sounds"

HONORS AND AWARDS

DST-INSPIRE Fellowship | Dept. of Science & Technology, Govt. of India

Jan 2021

- Prestigious national fellowship awarded to top PhD students across India for outstanding academic performance.

DBT-JRF (Category I) | Dept. of Biotechnology, Govt. of India

June 2020

- Awarded Category I rank under the DBT-Junior Research Fellowship program via the BET national entrance examination.

CONFERENCES AND WORKSHOPS

EMBO Workshop: Non-coding RNA Medicine

May 2023

- Attended an international EMBO workshop on RNA therapeutics and non-coding RNA biology held in Poznań, Poland.

Data Dynamics Summit (DDS 2024), IISER Pune

March 2024

- Participated in a national-level summit focusing on data science, computational biology, and interdisciplinary analytics.

IBAB Bio-IT Workshop: Introductory Omics and RNA-seq Data Analysis

July 2024

- Completed a 3-day hands-on workshop on RNA-seq analysis, omics data processing, and bioinformatics pipelines at IBAB, Bangalore.

Symposium on Computational Gastronomy, IIIT Delhi

Dec 2023

- Participated in an interdisciplinary symposium exploring computational approaches to food science and data-driven gastronomy.

Symposium on Computational Gastronomy, IIIT Delhi

Dec 2022

- Attended symposium sessions on computational modeling and data analysis in food systems.

ADDITIONAL INFORMATION

Languages: English (C1 – Advanced) · Hindi (B2 – Upper-Intermediate) · Odia (C2 – Native Proficiency)

REFERENCES

Prof. Gajendra P.S. Raghava
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Project Mentor (Master's Thesis)